

AWS storage and Content delivery Services-43

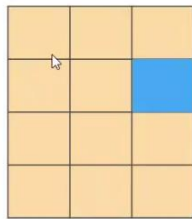
Introduction

- Storage and content delivery are core services in AWS.
- They provide scalable, durable, and cost-effective solutions to store, retrieve, and distribute data.
- AWS offers different storage types for various use cases, from object storage to block storage and file storage, along with global content delivery through Amazon CloudFront.

AWS storage options: Block storage versus object storage

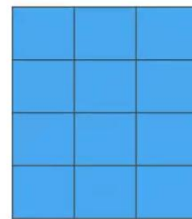


What if you want to change **one character** in a 1-GB file?



Block storage

Change one block (piece of the file)
that contains the character



Object storage

Entire file must be updated



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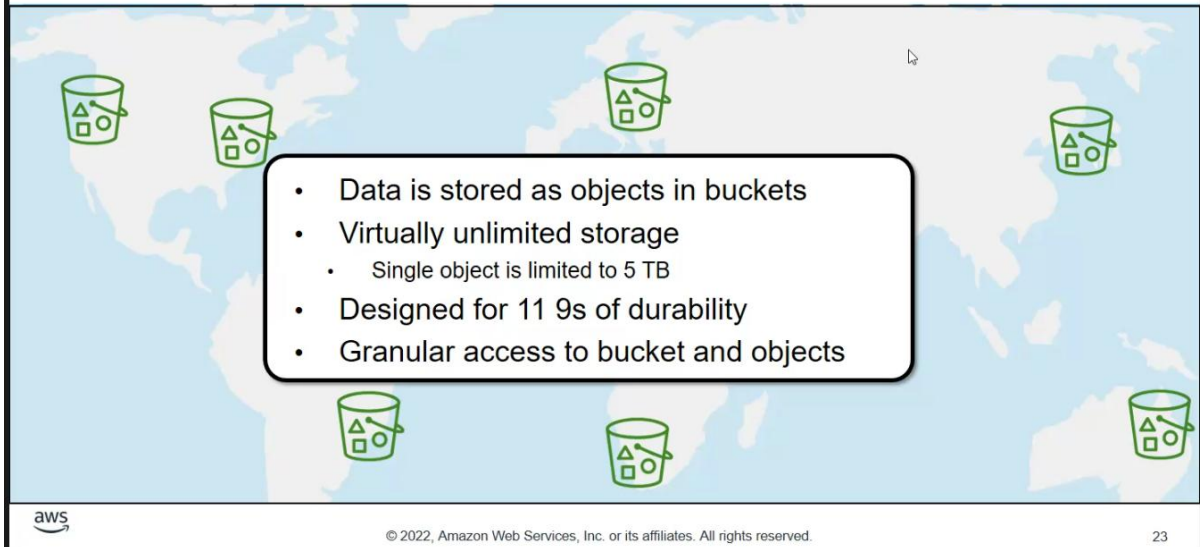
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AWS Storage Services

- **Amazon Simple Storage Service (Amazon S3)**
 - Object storage designed for durability.
 - Stores any type of data: documents, images, videos, backups.
 - Supports versioning, lifecycle policies, replication across regions.
 - Storage classes:
 - S3 Standard (frequent access)
 - S3 Intelligent-Tiering (automated cost optimization)
 - S3 Standard-IA (infrequent access)
 - S3 Glacier and Glacier Deep Archive (archival).

- Common use: application data storage, static website hosting, media storage.

Amazon S3 overview



- **Amazon Elastic Block Store (EBS)**

- Block-level storage used with EC2 instances.
- Provides persistent storage for virtual machines.
- Storage types:
 - General Purpose SSD (gp3)
 - Provisioned IOPS SSD (io2)
 - Throughput Optimized HDD (st1)
 - Cold HDD (sc1).
- Common use: databases, operating systems, applications needing low latency.

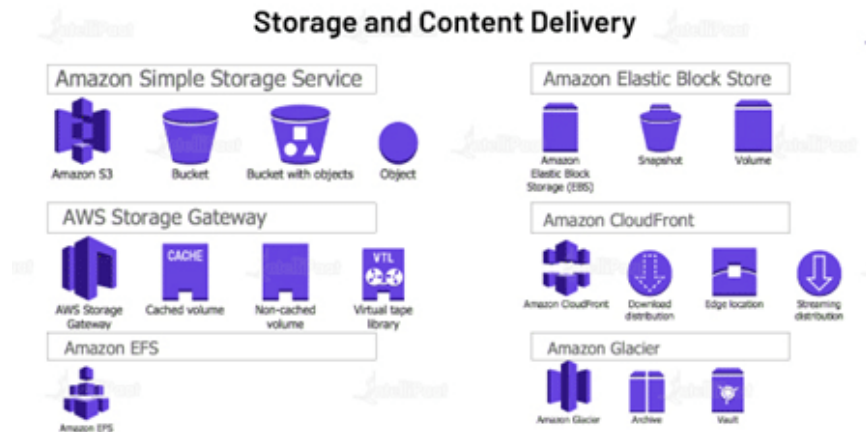
- **Amazon Elastic File System (EFS)**

- Scalable file storage that grows and shrinks automatically.
- Fully managed, serverless, and supports NFS protocol.
- Shared across multiple EC2 instances.
- Common use: big data analytics, content repositories, home directories.

- **Amazon FSx**

- Managed Windows and Linux file systems.
- FSx for Windows File Server integrates with Active Directory.
- FSx for Lustre provides high-performance file storage for HPC workloads.
- **Amazon Glacier and Glacier Deep Archive**
 - Low-cost archival storage.
 - Retrieval options range from minutes to hours.
 - Common use: compliance data, historical records, long-term backups.
- **AWS Storage Gateway**
 - Hybrid storage service that connects on-premises environments with AWS cloud storage.
 - Provides three modes:
 - File Gateway (NFS/SMB for cloud-backed file storage).
 - Volume Gateway (block storage with snapshots).
 - Tape Gateway (cloud-based tape storage for backups).
- **AWS Backup**
 - Centralized service to automate and manage backups across AWS services (EBS, RDS, DynamoDB, EFS).
 - Helps meet compliance requirements with scheduled policies.

AWS Content Delivery Services



- **Amazon CloudFront**
 - Content Delivery Network (CDN).
 - Delivers content (web pages, videos, APIs) globally with low latency.
 - Uses a network of **Edge Locations** distributed worldwide.
 - Integrates with S3, EC2, Elastic Load Balancing, and Lambda@Edge.
 - Security features: AWS Shield for DDoS protection, AWS WAF for firewall protection.
 - Common use: distributing static and dynamic web content, video streaming, software downloads.
- **AWS Global Accelerator**
 - Improves application availability and performance for global users.
 - Uses the AWS global network to route traffic to the nearest healthy endpoint.
 - Supports both TCP and UDP traffic.
- **Amazon Route 53**
 - Scalable Domain Name System (DNS) service.
 - Provides domain registration, DNS routing, and health checks.
 - Works with CloudFront to optimize delivery of web applications.

Example Use Case: Media Streaming Application

- Store media files (movies, images) in Amazon S3.

- Use S3 Glacier for archived media that is rarely accessed.
- Use EFS for shared file access among application servers.
- Distribute content globally using Amazon CloudFront.
- Route traffic with Route 53 for domain management.
- Improve speed for global viewers with AWS Global Accelerator.

Benefits of AWS Storage and Content Delivery

- Scalability: Storage automatically grows with demand.
- Cost-effectiveness: Pay only for what you use, choose storage tiers.
- Durability and reliability: Good durability in S3.
- Low latency: Edge locations deliver content closer to users.
- Security: Encryption, access control policies, compliance certifications.
- Flexibility: Multiple storage types for different workloads.

AWS storage services (S3, EBS, EFS, FSx, Glacier) provide robust and flexible storage for all workloads.

AWS content delivery services (CloudFront, Global Accelerator, Route 53) ensure fast and secure content distribution across the globe.

Together, they enable businesses to store, protect, and deliver data efficiently while ensuring performance and compliance.